

ESG Data Platform - Executive Summary

Executive Summary — A Personal Data Engineering Portfolio Project

Ishan Peshkar

July 2026

Enterprise ESG Data Platform

Executive Summary

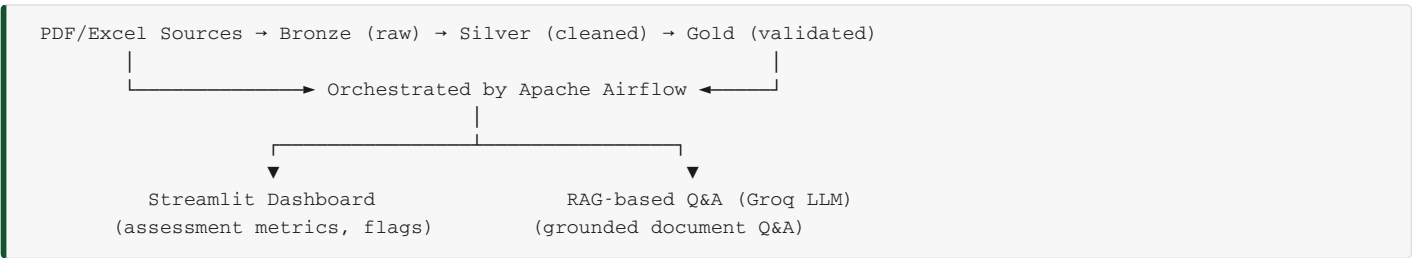
Background

While working as an intern on an ESG/EPD (Environmental Product Declaration) document analysis project — extracting data from sustainability documents and assessing it against regulatory rulesets — I wanted to deepen my Data Engineering skills through a parallel personal project. Rather than build a generic pipeline, I chose to build a scaled-down, fully local, and completely original version of the kind of ESG data platform that firms like Deloitte, EY, and Infosys build for enterprise clients.

Goal: learn real Data Engineering practices (extraction, validation, orchestration, data modeling) while applying my existing AI/GenAI/RAG background to a domain I already understood — without using any real company data, and without any paid tools or cloud subscriptions.

What Was Built

A complete, local, end-to-end data platform processing real public ESG/EPD documents:



Core components:

- **Extraction** — PDF (pdfplumber) and Excel (pandas) parsers producing raw “bronze” records
- **Cleaning** — rule-based field extraction into structured “silver” tables (DuckDB)
- **Validation** — a two-layer validation system: structural schema checks (Pandera) and domain-specific business rules, producing a “gold” layer split into valid vs. flagged records
- **Orchestration** — the full pipeline runs as a single Apache Airflow DAG (containerized via Docker/WSL2), replacing manual script execution with a scheduled, dependency-aware, retryable pipeline
- **Dashboard** — a Streamlit application surfacing pass/fail metrics and flagged-record detail, mirroring a real compliance-assessment interface
- **AI-Powered Q&A (RAG)** — documents are chunked, embedded locally (sentence-transformers), and stored in a vector database (ChromaDB); user questions are answered by an LLM (Groq) strictly grounded in retrieved document content, with source citations

Key Engineering Insights

This project intentionally treated failures as data, not noise. A few findings stood out:

1. **Syntactic validation ≠ semantic validation.** A schema can confirm a field is “shaped” correctly (e.g. a valid number) while still being factually wrong. All 8 documents passed structural validation, yet business-rule checks correctly flagged 100% of them as semantically unreliable — proving the validation system worked exactly as intended, even though the data wasn’t yet trustworthy.

- 2. **Regex-based extraction has a hard ceiling.** Naive keyword/pattern extraction worked for narrow, consistent fields (e.g. standard references) but failed on fields where real-world documents vary by issuer and layout (e.g. product names). This is direct, reproducible evidence for why production ESG platforms — including the one at my internship — rely on LLM-based, context-aware extraction instead.
- 3. **Idempotency matters.** An early version of the pipeline silently duplicated data on every re-run (24 records instead of 8) because bronze files were timestamped rather than keyed by source. Fixing this was a concrete, practical lesson in a core Data Engineering principle: pipelines should produce the same result no matter how many times they run.
- 4. **Grounded RAG correctly refuses to guess.** When asked out-of-scope questions (e.g. about a company never present in the ingested documents), the system explicitly stated the answer wasn't present in its retrieved context rather than falling back on the underlying LLM's general knowledge — the core discipline that separates a real RAG system from simply calling an LLM directly.

Tech Stack (100% local, zero cost)

Layer	Tool
Extraction	pdfplumber, pandas
Storage	DuckDB (silver & gold layers)
Validation	Pandera (schema) + custom business rules
Orchestration	Apache Airflow (Docker + WSL2)
Dashboard	Streamlit
Embeddings	sentence-transformers (all-MiniLM-L6-v2)
Vector store	ChromaDB
LLM	Groq (Llama 3.3 70B, free tier)

Status & Next Steps

All five planned phases (Foundations → Manual Pipeline → Validation → Orchestration → Dashboard/RAG) are complete and verified end-to-end. Potential future extensions include scheduled/automated DAG runs, additional source types (Excel silver-layer cleaning, government APIs), and swapping the current rule-based PDF field extraction for the LLM-based structured extraction validated as necessary in Phase 1.

For the full technical build log — every decision, command, bug, and fix — see the companion document: **Technical Deep Dive**.